

FORMAL LANGUAGE AND AUTOMATA THEORY

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS12	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Formal languages and automata theory deals with the concepts of automata, formal languages, grammar, computability and decidability. The reasons to study Formal Languages and Automata Theory are Automata Theory provides a simple, elegant view of the complex machine that we call a computer. Automata Theory possesses a high degree of permanence and stability, in contrast with the ever-changing paradigms of the technology, development, and management of computer systems. Further, parts of the Automata theory have direct bearing on practice, such as Automata on circuit design, compiler design, and search algorithms; Formal Languages and Grammars on compiler design; and Complexity on cryptography and optimization problems in manufacturing, business, and management. Last, but not least, research oriented students will make good use of the Automata theory studied in this course.

COURSE OBJECTIVES:

1. To present the theory of finite automata as the first step towards learning advanced topics such as compiler design.
2. To discuss the applications of finite automata towards text processing.
3. To develop an understanding of Regular expressions and context free grammars and how these concepts are used in lexical analyzer.
4. To develop an understanding of finite automata through Turing machines

COURSE OUTCOMES:

At the end of the course students will be able to:

1. To use basic concepts of formal languages of finite automata techniques
2. To Design Finite Automata's for different Regular Expressions and Languages
3. To Construct context free grammar for various languages
4. To solve various problems of applying normal form techniques, push down automata and Turing Machines.

UNIT-I	INTRODUCTION:FINITE AUTOMATA	
FINITE AUTOMATA (FA): Introduction, Deterministic Finite Automata (DFA) -Formal definition, simpler notations (state transition diagram, transition table), language of a DFA. Nondeterministic Finite Automata (NFA)- Definition of NFA, language of an NFA, Equivalence of Deterministic and Nondeterministic Finite Automata, Applications of Finite Automata, Finite Automata with Epsilon Transitions, Eliminating Epsilon transitions, Minimization of Deterministic Finite Automata, Finite automata with output (Moore and Mealy machines) and Inter conversion.		
UNIT-II	REGULAR EXPRESSIONS (RE)	
REGULAR EXPRESSIONS (RE): Introduction, Identities of Regular Expressions, Finite Automata and Regular Expressions- Converting from DFA to Regular Expressions, Converting Regular Expressions to Automata, applications of Regular Expressions. REGULAR GRAMMARS: Definition, regular grammars and FA, FA for regular grammar, Regular grammar for FA. Proving languages to be non-regular - Pumping lemma, applications, Closure properties of regular languages.		
UNIT-III	CONTEXT FREE GRAMMER (CFG):	

CONTEXT FREE GRAMMER (CFG): Derivation Trees, Sentential Forms, Rightmost and Leftmost derivations of Strings. AMBIGUITY IN CFG: Elimination of ambiguity, Minimization of CFG's, CNF, GNF, Pumping Lemma for CFL's,		
UNIT-IV	PUSHDOWN AUTOMATA	
PUSHDOWN AUTOMATA: Definition, Model, Acceptance of CFL, Acceptance by Final State and Acceptance by Empty stack and its Equivalence, Equivalence of CFG and PDA. TURING MACHINES (TM): Formal definition and behavior, Languages of a TM, TM as accepters, and TM as a computer of integer functions, Types of TM's.		
UNIT-V	RECURSIVELY ENUMERABLE LANGUAGES	
RECURSIVELY ENUMERABLE LANGUAGES (REL): Properties of Recursively Enumerable Languages, Universal Turing machine, The Halting problem, Context sensitive language and linear bounded automata (LBA), Chomsky hierarchy, Decidability, Post's correspondence problem (PCP), Undecidability of PCP.		
TEXT BOOKS:		
1. John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman (2007), Introduction to Automata Theory Languages and Computation, 3rd Edition, Pearson Education, India 2. Theory of Computer Science (Automata, Languages and Computation), 2nd Edition by K. L. P. Mishra and N. Chandrasekharan, PHI.		
REFERENCE BOOKS:		
1. Finite Automata and Formal Languages: A Simple Approach, A. M. Padma Reddy, Pearson Education India. 2. An Introduction to FORMAL LANGUAGES and AUTOMATA (Fifth Edition), PETER LINZ.		
WEB REFERENCES:		
1. https://nptel.ac.in/courses/111103016/ 2. https://www.ics.uci.edu/~goodrich/teach/cs162/notes/ 3. https://www.geeksforgeeks.org/theory-of-computation-automata-tutorials/		
E-Text Books:		
1. https://mcdtu.files.wordpress.com/2017/03/introduction-to-automata-theory.pdf 2. https://www.amazon.in/Formal-Languages-Automata-Theory-agpal/dp/019807106X		
MOOC COURSE		
1. https://swayam.gov.in/course/3831-theory-of-computation 2. https://www.udemy.com/theory-of-automata/ 3. https://nptel.ac.in/courses/111103016/		